

Transition to Electrostatic Spray Technology in Beverage Bottle Production

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Abstract—This report presents a transition-to-production study for replacing the existing air-spray beverage bottle coating process with an automated electrostatic spray system. The main objective is to reduce paint consumption, improve coating quality, increase throughput, reduce unplanned downtime, and evaluate the financial feasibility of full-scale implementation. The study uses requirements translation, process-flow analysis, discrete-event simulation planning, predictive maintenance analysis, and cost-benefit evaluation. Based on the collected production data, the existing line produces approximately 90,000 bottles per day and consumes about 15,000 L of paint per month. The proposed electrostatic spray system is expected to increase throughput to 108,000 bottles per day, reduce paint use by approximately 19–20%, reduce annual downtime by 40%, and improve overall equipment effectiveness from 80% to 91%. Using an exchange rate assumption of 1 USD = 94.61 INR, the installed capital cost of the spray system is approximately \$22,196, while the estimated total annual benefit is approximately \$168,058. The payback period is approximately 1.6 months under the target savings model and 1.84 months under the measured site-trial ROI data.

Index Terms—Electrostatic spray, beverage bottle production, transition to production, discrete-event simulation, predictive maintenance, ROI, OEE, smart manufacturing.

I. INTRODUCTION

Beverage bottle coating is a high-volume manufacturing process where coating quality, paint consumption, equipment availability, and line throughput directly affect production cost and customer satisfaction. The existing process uses air-spray guns mounted near a conveyor-based bottle/can line. Although this setup is functional, it has important limitations, including high paint usage, overspray waste, refill interruptions, daily cleaning downtime, quality variation, and maintenance-related stoppages.

This project studies the transition from the baseline air-spray process to an automated electrostatic spray technology for beverage bottle production. Electrostatic spray coating charges paint particles before deposition, improving transfer efficiency and reducing overspray. The proposed system uses automated electrostatic guns mounted around the conveyor to achieve more uniform coating coverage while reducing material loss.

The project is not focused on physically building the equipment. Instead, it analyzes the transition from design to production using a systems engineering approach. The study includes customer requirements, system requirements, process architecture, simulation metrics, predictive maintenance

improvement, financial justification, risk analysis, and asset-management integration.

II. PROBLEM STATEMENT

The current coating line operates at approximately 90,000 bottles per day with high paint consumption and recurring operational interruptions. The production data indicates monthly paint consumption of approximately 15,000 L, with paint cost assumed at 400 INR/L, equivalent to approximately \$4.23/L. The existing line also requires daily cleaning and periodic deep cleaning, and it experiences unplanned downtime due to equipment failures, clogging, refill delays, and manual intervention.

The main problem is to determine whether electrostatic spray technology can be transitioned into production in a technically feasible and financially justified way. The study evaluates whether the proposed system can:

- reduce paint consumption and coating waste,
- increase daily production throughput,
- reduce downtime and maintenance interruptions,
- improve coating quality and first-pass yield,
- achieve fast payback and strong return on investment.

III. CURRENT AND PROPOSED PRODUCTION SYSTEM

The current production flow consists of in-house production, furnace operation, paint booths, quality inspection, and packaging. Bottles or cans move on a conveyor through the coating area. The baseline process uses conventional air-spray guns, while the proposed process replaces them with electrostatic spray guns and continuous paint-feed support.

A. Baseline Process Flow

The simplified baseline production flow is:

Bottle Generator → Spray Queue → Spray Booth (1)

→ Curing Oven

→ Inspection → Packaging.(2)

The baseline system includes manual refill delays, lower transfer efficiency, higher defect rate, and higher maintenance downtime.

B. Proposed Electrostatic Process Flow

The proposed new line is:

Bottle Generator → ES Spray Queue (3)

→ Electrostatic Spray Booth → Curing Oven (4)

→ Inspection → Packaging. (5)

The electrostatic system uses automated guns mounted around the conveyor for consistent 360-degree coating coverage. The expected improvements come from higher paint transfer efficiency, faster spray service time, fewer refill interruptions, improved reliability, and reduced defect rate.

IV. CUSTOMER REQUIREMENTS AND SYSTEM REQUIREMENTS

The transition study begins by translating customer needs into measurable engineering requirements. Table I summarizes the key requirement mapping.

TABLE I
CUSTOMER REQUIREMENTS TO SYSTEM REQUIREMENTS MAPPING

Customer Requirement	System Requirement
Reduce paint consumption	Achieve approximately 19–20% paint savings compared with air-spray coating.
Improve coating quality	Reduce defect rate from 3% to approximately 1%.
Increase throughput	Increase daily output from 90,000 to 108,000 bottles/day.
Reduce downtime	Reduce unplanned downtime from 500 to 300 hr/year.
Improve reliability	Increase MTBF from 4,000 hr to 6,000 hr.
Reduce repair time	Reduce MTTR from 20 hr to 12 hr.
Improve sustainability	Reduce overspray, paint waste, and VOC-related material loss.
Improve production control	Integrate OEE monitoring and maintenance tracking dashboard.
Maintain conveyor compatibility	Keep conveyor speed approximately 35 mm/s.
Achieve fast payback	Recover capital cost in less than 2 months.

V. DATA COLLECTION AND BASELINE PARAMETERS

The project uses production data collected from the existing beverage bottle coating operation and projected parameters for the proposed electrostatic line. The main baseline and proposed values are shown in Table II.

VI. DISCRETE-EVENT SIMULATION METHOD

A discrete-event simulation model is proposed using Jaam-Sim to compare the baseline air-spray line against the electrostatic spray line. The simulation entities are bottles, and the main process objects are generator, queue, spray booth, curing oven, maintenance logic, and sink.

TABLE II
BASELINE AND PROPOSED SYSTEM PARAMETERS

Parameter	Baseline	Electrostatic	Unit
Production rate	90,000	108,000	bottles/day
Inter-arrival time	0.96	0.80	s
Service time	0.90	0.72	s/bottle
Conveyor speed	35	35	mm/s
MTBF	4,000	6,000	hr
MTTR	20	12	hr
Downtime	500	300	hr/year
Paint transfer efficiency	70	88	%
Paint used/day	600	480	L/day
Paint used/month	15,000	12,000	L/month
Defect rate	3	1	%
Refill delay	30 s/1000 bottles	None	–
OEE	80	91	%

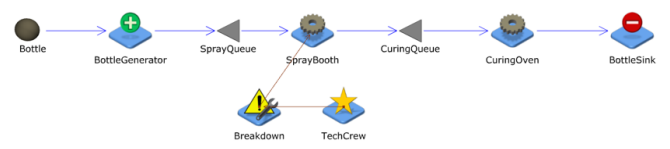


Fig. 1. Simulation Model

A. Simulation Metrics

The main simulation metrics are throughput, cycle time, equipment utilization, bottleneck location, downtime, and OEE.

Throughput is calculated as:

$$Q = \frac{N_{\text{completed}}}{T_{\text{days}}}, \quad (6)$$

where $N_{\text{completed}}$ is the number of bottles completed and T_{days} is the simulated time in days.

Cycle time is calculated as:

$$CT = \frac{1}{N} \sum_{i=1}^N (t_{\text{completion},i} - t_{\text{arrival},i}). \quad (7)$$

Equipment utilization is calculated as:

$$U = \frac{T_{\text{busy}}}{T_{\text{available}} \times n_{\text{guns}}} \times 100. \quad (8)$$

For example, if one gun is busy for 432,000 seconds during a 7-day run:

$$T_{\text{available}} = 7 \times 24 \times 3600 = 604,800 \text{ s}, \quad (9)$$

$$U = \frac{432,000}{604,800} \times 100 = 71.43\%. \quad (10)$$

B. Simulation Scenarios

Three simulation cases are recommended:

- 1) **Baseline Case:** air-spray guns, 0.96 s inter-arrival time, 0.90 s service time, MTBF = 4,000 hr, MTTR = 20 hr, and 30 s refill delay every 1,000 bottles.

- 2) **Electrostatic Case:** electrostatic guns, 0.80 s inter-arrival time, 0.72 s service time, MTBF = 6,000 hr, MTTR = 12 hr, and no refill delay.
- 3) **Sensitivity Case:** service time improvement varied from 10% to 30%, MTBF varied from 4,000 to 7,000 hr, and number of guns varied from 2 to 4.

VII. SIMULATION RESULTS

The expected simulation results are summarized in Table III. These values represent the target performance after transitioning to electrostatic spray technology.

TABLE III
BASELINE VS. ELECTROSTATIC SIMULATION RESULTS

Metric	Baseline	Electrostatic	Change
Throughput	90,000	108,000	+20%
Avg. cycle time	0.96 s	0.80 s	-17%
Service time	0.90 s	0.72 s	-20%
Utilization	82%	90%	+8%
Downtime	500 hr/yr	300 hr/yr	-40%
Transfer efficiency	70%	88%	+18 points
Paint/month	15,000 L	12,000 L	-20%
Defect rate	3%	1%	-66.7%
OEE	80%	91%	+11 points

The production increase is:

$$\Delta Q = 108,000 - 90,000 = 18,000 \text{ bottles/day.} \quad (11)$$

The percentage throughput improvement is:

$$\% \Delta Q = \frac{18,000}{90,000} \times 100 = 20\%. \quad (12)$$

The cycle time reduction is:

$$\% \Delta CT = \frac{0.96 - 0.80}{0.96} \times 100 = 16.67\% \approx 17\%. \quad (13)$$

The defect reduction is:

$$\% \Delta D = \frac{3 - 1}{3} \times 100 = 66.67\%. \quad (14)$$

VIII. PREDICTIVE MAINTENANCE ANALYSIS

Predictive maintenance is included to reduce unplanned downtime and improve equipment availability. The baseline line experiences approximately 500 hr/year of unplanned downtime. With improved monitoring, sensors, and condition-based maintenance, the new system is projected to reduce downtime to 300 hr/year.

The downtime reduction is:

$$\Delta D_t = 500 - 300 = 200 \text{ hr/year.} \quad (15)$$

The percentage downtime reduction is:

$$\% \Delta D_t = \frac{200}{500} \times 100 = 40\%. \quad (16)$$

The production rate per hour is:

$$R_h = \frac{90,000}{24} = 3,750 \text{ bottles/hr.} \quad (17)$$

The recovered annual production is:

$$Q_{\text{recovered}} = 200 \times 3,750 = 750,000 \text{ bottles/year.} \quad (18)$$

Assuming a contribution margin of 2 INR/bottle:

$$B_{\text{PdM,INR}} = 750,000 \times 2 = 1,500,000 \text{ INR/year.} \quad (19)$$

Using 1 USD = 94.61 INR:

$$B_{\text{PdM,USD}} = \frac{1,500,000}{94.61} = \$15,855/\text{year.} \quad (20)$$

TABLE IV
PREDICTIVE MAINTENANCE IMPACT

Metric	Baseline	New Line	Impact
Unplanned downtime	500 hr/yr	300 hr/yr	-40%
Bottles lost	1.875M	1.125M	0.75M recovered
Additional value	–	\$15,855/yr	production gain
Availability	94%	97%	+3 points

IX. COST-BENEFIT AND ROI ANALYSIS

The economic analysis includes capital cost, monthly paint savings, annual material savings, predictive maintenance gains, total annual benefit, ROI, and payback period. All currency values are converted using:

$$1 \text{ USD} = 94.61 \text{ INR.} \quad (21)$$

A. Paint Cost

The paint price is:

$$C_p = 400 \text{ INR/L} = \frac{400}{94.61} = \$4.23/\text{L.} \quad (22)$$

Baseline monthly paint cost:

$$C_{\text{base}} = 15,000 \times 400 = 6,000,000 \text{ INR/month.} \quad (23)$$

$$C_{\text{base,USD}} = \frac{6,000,000}{94.61} = \$63,418/\text{month.} \quad (24)$$

New monthly paint cost under the 20% target case:

$$C_{\text{new}} = 12,000 \times 400 = 4,800,000 \text{ INR/month.} \quad (25)$$

$$C_{\text{new,USD}} = \frac{4,800,000}{94.61} = \$50,735/\text{month.} \quad (26)$$

Monthly material savings:

$$S_{\text{month}} = 6,000,000 - 4,800,000 = 1,200,000 \text{ INR/month.} \quad (27)$$

$$S_{\text{month,USD}} = \frac{1,200,000}{94.61} = \$12,684/\text{month.} \quad (28)$$

Annual material savings:

$$S_{\text{annual}} = 1,200,000 \times 12 = 14,400,000 \text{ INR/year.} \quad (29)$$

$$S_{\text{annual,USD}} = \frac{14,400,000}{94.61} = \$152,204/\text{year.} \quad (30)$$

B. Capital Cost

The installed package cost for the electrostatic spray system, including 3 guns and accessories, is:

$$CAPEX = 2,100,000 \text{ INR.} \quad (31)$$

$$CAPEX_{USD} = \frac{2,100,000}{94.61} = \$22,196. \quad (32)$$

C. Total Annual Benefit

The total annual benefit combines material savings and predictive maintenance production gain:

$$B_{\text{total,INR}} = 14,400,000 + 1,500,000 = 15,900,000 \text{ INR/year.} \quad (33)$$

$$B_{\text{total,USD}} = \frac{15,900,000}{94.61} = \$168,058/\text{year.} \quad (34)$$

D. Payback Period

Monthly benefit is:

$$B_{\text{month}} = \frac{15,900,000}{12} = 1,325,000 \text{ INR/month.} \quad (35)$$

The payback period is:

$$P = \frac{2,100,000}{1,325,000} = 1.58 \text{ months} \approx 1.6 \text{ months.} \quad (36)$$

E. Annual ROI

Annual ROI is calculated as:

$$ROI = \frac{B_{\text{total}}}{CAPEX} \times 100. \quad (37)$$

$$ROI = \frac{15,900,000}{2,100,000} \times 100 = 757.1\% \approx 750\%. \quad (38)$$

TABLE V
COST-BENEFIT SUMMARY IN USD

Category	Value
Paint price	\$4.23/L
Baseline paint cost	\$63,418/month
New paint cost	\$50,735/month
Monthly material savings	\$12,684/month
Annual material savings	\$152,204/year
PdM production gain	\$15,855/year
Total annual benefit	\$168,058/year
Installed CAPEX	\$22,196
Payback period	1.6 months
Annual ROI	757%

F. Measured Trial Savings Check

The site trial showed that 2 L of paint coated 131 bottles with the existing air-spray gun and 156 bottles with the electrostatic gun. The paint-efficiency improvement is:

$$\%S = \frac{156 - 131}{131} \times 100 = 19.08\% \approx 19\%. \quad (39)$$

Using the measured 19% saving, the material saving is approximately 1,140,000 INR/month:

$$S_{19\%,USD} = \frac{1,140,000}{94.61} = \$12,049/\text{month.} \quad (40)$$

Annual measured material savings are:

$$S_{19\%,\text{annual}} = 13,680,000 \text{ INR/year} = \$144,594/\text{year.} \quad (41)$$

This explains why the site-trial ROI payback is reported as approximately 1.84 months, while the rounded target financial model gives approximately 1.6 months after including predictive maintenance output gain.

X. RISK ANALYSIS AND MITIGATION

Technology transition introduces technical, operational, financial, and data-related risks. Table VI summarizes the key risks and mitigation actions.

TABLE VI
RISK ANALYSIS AND MITIGATION PLAN

Risk	Mitigation Strategy
Inconsistent baseline data	Use standard data templates, Gemba observations, and cross-checks between production, maintenance, and quality records.
Incorrect MTBF/MTTR assumptions	Use historical downtime logs, industry benchmarks, and monthly reliability updates after pilot deployment.
Integration issues	Validate gun mounting, grounding, safety interlocks, airflow, paint feed, and conveyor compatibility before full-scale rollout.
Operator resistance	Provide training on safety, cleaning, startup, shutdown, and troubleshooting.
Post-implementation maintenance issues	Use preventive maintenance, vendor support, spare-parts planning, and condition monitoring.
Financial overestimation	Report both measured 19% savings and target 20% savings; perform sensitivity analysis.
Safety and compliance	Verify electrical grounding, overspray control, ventilation, and environmental compliance.

XI. ASSET MANAGEMENT INTEGRATION

The electrostatic spray transition should be supported by a digital asset-management strategy. The following elements are recommended:

- **Predictive maintenance system:** Install sensors to monitor voltage, pressure, flow, spray hours, clogging behavior, and compressor performance.

- **Centralized asset registry:** Track spray guns, hoses, maintenance schedules, spare parts, service records, and calibration history.
- **Condition-based maintenance:** Trigger maintenance using real-time operating data instead of fixed intervals only.
- **OEE dashboard:** Track availability, performance, quality, throughput, downtime, defect rate, and paint consumption by shift.
- **Lifecycle cost management:** Monitor paint usage, energy cost, maintenance cost, spare-part cost, and total cost of ownership.

XII. IMPLEMENTATION PLAN

The recommended transition plan follows a phased approach:

- 1) **Weeks 1–2: Baseline study.** Collect paint usage, throughput, downtime, defect, cleaning, and refill data.
- 2) **Weeks 3–4: Process modeling.** Build baseline and electrostatic process models in JaamSim.
- 3) **Weeks 5–6: Reliability analysis.** Estimate MTBF, MTTR, downtime reduction, and predictive maintenance benefits.
- 4) **Weeks 7–8: Financial analysis.** Calculate material savings, production gains, payback, and ROI.
- 5) **Week 9+: Production planning.** Finalize documentation, review risks, train operators, and plan pilot/full-scale deployment.

XIII. DISCUSSION

The results show that electrostatic spray technology has strong technical and financial justification for beverage bottle coating. The main operational benefit is the increase in paint transfer efficiency from 70% to 88%, which reduces overspray and material waste. The process also improves throughput by reducing service time from 0.90 s to 0.72 s per bottle and reducing inter-arrival time from 0.96 s to 0.80 s. These changes support a 20% output increase from 90,000 to 108,000 bottles per day.

The maintenance benefit is also significant. By improving MTBF from 4,000 hr to 6,000 hr and reducing MTTR from 20 hr to 12 hr, the expected annual downtime decreases from 500 hr to 300 hr. This recovers approximately 750,000 bottles per year and contributes approximately \$15,855 in additional annual production value.

Financially, the system is attractive because the installed cost is approximately \$22,196, while the total annual benefit is approximately \$168,058. Even if only the measured 19% paint saving is used, the project still demonstrates a payback period below two months. This makes the transition suitable for full-scale implementation after pilot validation.

XIV. CONCLUSION

This project analyzed the transition from conventional air-spray coating to electrostatic spray technology in beverage

bottle production. The proposed system improves throughput, coating quality, paint efficiency, reliability, and cost performance. The expected output increases from 90,000 to 108,000 bottles per day, while paint consumption decreases from 15,000 L/month to approximately 12,000 L/month under the target model. The measured site-trial savings are approximately 19%, and the planning target is approximately 20%. Predictive maintenance reduces unplanned downtime by 40%, improving annual production capacity and equipment availability.

Using 1 USD = 94.61 INR, the installed electrostatic system cost is approximately \$22,196. The estimated total annual benefit is approximately \$168,058/year, with a payback period of approximately 1.6 months and annual ROI of approximately 757%. Based on the technical performance, maintenance improvement, and financial return, the electrostatic spray transition is feasible, scalable, and strongly justified for production deployment.